

Bonus: Movie Watchlist with `std::vector`

Write a C++ program that manages a small movie watchlist.

Learning goals

- Create and use a `vector<string>`
- Add elements with `push_back`
- Insert one element with `insert`
- Remove one element with `erase`
- Traverse a vector with an `iterator`
- Use `size`, `capacity`, and `reserve`

Task

Write a program that does the following:

1. Create an empty `vector<string>` called `watchlist`.
2. Reserve space for 10 movies.
3. Add these movies using `push_back`:
 - "Thomas Becket"
 - "The Island"
 - "Pride and Prejudice"
 - "The Reluctant Saint"
4. Insert "The Robe" before "The Reluctant Saint".
5. Remove "Pride and Prejudice" from the list.
6. Print the final watchlist using an `iterator`.
7. Print the vector `size` and `capacity`.

Expected output

Movie watchlist:
1. Thomas Becket
2. The Island
3. The Robe
4. The Reluctant Saint

Size: 4
Capacity: 10

Starter code

```
#include <iostream>
#include <vector>
#include <string>
using namespace std;

int main()
{
    // Create empty vector

    // Reserve space for 10 movies

    // Add four movies

    // Insert "The Robe" before "The Reluctant Saint"

    // Remove "Pride and Prejudice"

    // Print final watchlist using an iterator

    // Print size and capacity

    return 0;
}
```

Notes for checking

- The program should compile without warnings or errors.

- The final vector should contain exactly 4 movies.
- The capacity should be at least 10, and normally exactly 10 after `reserve(10)`.

Submission

Send me the URL to your solution in [OneCompiler.com](https://onecompiler.com) via email.